

ANIRUDH ADDA

Data Engineer | Multi-Cloud

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PROFESSIONAL SUMMARY

Data Engineer with 4+ years of experience building production-grade data infrastructure for regulated financial services environments. Specialized in designing scalable ETL/ELT pipelines, real-time streaming architectures, and cloud-native data platforms across AWS, GCP, and Azure. Track record of reducing pipeline latency, automating infrastructure provisioning, and enabling analytics at enterprise scale. Seeking to apply multi-cloud data engineering expertise to support AI/ML infrastructure and research data systems.

TECHNICAL SKILLS

Cloud Platforms	AWS (S3, Glue, Lambda, Redshift, Kinesis, Step Functions) · GCP (BigQuery, Dataflow, Pub/Sub, Cloud Composer, Dataproc) · Azure (Data Lake, Synapse Analytics, ADF) · Databricks
Languages	Python · SQL · PySpark · Bash
Data Engineering	Apache Airflow · Apache Spark · Apache Kafka · Apache Beam · Apache Flink · Hadoop / HDFS · Hive
Databases	PostgreSQL · MySQL · MongoDB · Cassandra · Cloud SQL · HBase
Data Warehousing	Snowflake · Amazon Redshift · Google BigQuery · Azure Synapse · Dimensional Modeling · Star Schema · OLAP
DevOps & IaC	Terraform · Docker · Kubernetes · Jenkins · GitHub Actions · CI/CD Pipelines · CloudFormation
Monitoring	AWS CloudWatch · Grafana · Prometheus · Airflow Monitoring · Logging & Alerting
BI & Visualization	Power BI · Tableau · Looker
AI / Data Projects	ChromaDB · LLM Integration (Llama 3) · RAG Pipelines · FastAPI · Streamlit · dbt · SCD Type 2 · Kimball Modeling · Random Forest · Great Expectations · Data Lineage

WORK EXPERIENCE

U.S. Bank National Association — Data Engineer | Atlanta, GA Jul 2025 – Present

- Architected and owns 14 production AWS Glue + Redshift pipelines ingesting ~800 GB/day of transactional data, reducing regulatory reporting latency from D+2 to same-day (D+0) for 9 downstream teams.
- Built real-time Kafka + Spark Streaming ingestion layer processing 120K+ transactions/hour for fraud detection and operational dashboards, cutting event-to-insight latency from 4 hours to under 10 minutes.
- Optimized 3 high-cost PySpark jobs (Pandas + PySpark rewrites), reducing average runtime from 3.5 hrs to 45 minutes and AWS compute spend by ~\$22K/year.
- Automated infrastructure provisioning with Terraform and containerized pipeline deployments via Docker + Kubernetes, reducing deployment cycle time from 2 days to under 2 hours.

Reinsurance Group of America — Data Engineer | St. Louis, MO Aug 2024 – May 2025
(OPT/CPT — concurrent with Master's program)

- Designed and delivered 8 batch + streaming pipelines on Google Cloud Dataflow and Pub/Sub, reducing actuarial data processing time by ~45% and enabling weekly reporting to become daily.
- Rebuilt legacy ELT workflows in BigQuery and Cloud Data Fusion, improved query performance by 3× through partitioning and clustering optimizations, saving ~\$18K/year in BigQuery compute costs.
- Automated cloud infrastructure with Terraform and Cloud Deployment Manager, achieving 100% reproducible environment deployments and eliminating 6+ hrs/week of manual provisioning work.
- Built 4 executive-facing Power BI and Looker dashboards integrating BigQuery and Cloud SQL data, reducing ad-hoc data requests to the engineering team by ~30%.

Citco — Data Engineer | Hyderabad, India

Aug 2021 – Jul 2023

- Built enterprise data pipelines using Apache Kafka, Apache Spark, and Azure Data Factory to centralize operational data from 12 source systems into Azure Data Lake, onboarding 5 TB of historical data.
- Developed and optimized ETL workflows in Azure Synapse Analytics using Python and PySpark, processing 500 GB+ of daily transactional data with 99.6% SLA compliance over 18 months.
- Established CI/CD pipelines using Jenkins and GitHub Actions with Docker + Kubernetes deployments, reducing release cycle time from 1 week to 1 day and eliminating manual deployment errors.
- Delivered 6 Power BI and Tableau dashboards visualizing fund KPIs across 20+ stakeholders, replacing manual Excel-based reporting and saving ~8 hrs/week of analyst time.

EDUCATION

University of Dayton — M.S. Computer Science | Dayton, OH

Aug 2023 – May 2025

Relevant Coursework: Distributed Systems, Cloud Computing, Database Systems, Data Mining, Machine Learning Foundations

PERSONAL PROJECTS

Flight Intelligence ML Pipeline Python · Random Forest · Llama 3 · FastAPI · Streamlit · Airflow · Docker

- Built end-to-end production-grade ML pipeline ingesting 400,000+ flight records from 3 sources with automated conflict resolution and full custom data lineage tracking across every transformation stage.
- Implemented Great Expectations quality gates to validate data integrity before downstream processing, trained a self-retraining Random Forest model achieving 96.76% price prediction accuracy.
- Served predictions and insights through FastAPI, a 7-page interactive Streamlit dashboard, and an AI flight assistant powered by a locally hosted Llama 3 LLM, fully orchestrated with Airflow and containerized with Docker.

AI-Powered Financial Intelligence Pipeline Python · Llama 3 · ChromaDB · FastAPI · Airflow · Docker

- Built end-to-end RAG pipeline ingesting SEC filings from the EDGAR API, embedding documents into ChromaDB, and serving Q&A, summarization, and company-comparison queries using a locally hosted Llama 3 LLM.
- Built FastAPI backend and Streamlit front-end for real-time financial Q&A with semantic search across 500+ filings, implemented context-aware retrieval with relevance scoring.
- Orchestrated full pipeline with Airflow DAGs for scheduled ingestion and automated re-embedding, containerized end-to-end with Docker Compose for reproducible deployment.

Enterprise ETL Pipeline with Kimball Dimensional Modeling MySQL · dbt · Airflow · Power BI

- Designed a full 4-layer warehouse architecture (Staging → ODS → Star Schema → Data Mart) in MySQL following Kimball dimensional modeling best practices, implemented SCD Type 2 for complete historical tracking.
- Built modular, version-controlled dbt transformations with idempotent incremental load patterns, automated data quality auditing with row-count, null, and referential integrity checks at each layer.
- Implemented 3 distinct Airflow DAG patterns (full-refresh, incremental, and audit) with dependency management, retry logic, and pipeline health alerts, visualized KPIs in Power BI.

Automated Stock Market Batch Data Pipeline Python · PostgreSQL · Airflow · Docker · Streamlit

- Built fully automated batch pipeline pulling daily stock market data via public API, transforming it in Python, and loading to PostgreSQL, orchestrated with Airflow and visualized through an interactive Streamlit dashboard.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- AWS Certified Data Engineer – Associate